

Corentin Sautier

PhD Student in Computer Vision

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Education

- 2022–Today **PhD student**, *valeo.ai & ENPC, Imagine*, Paris.
Self-Supervised learning for sparse 3D point clouds using multimodal signals
Detailed achievements:
 - Developed better self-supervised methods on LiDAR point clouds.
 - Reached state-of-the-art LiDAR panoptic segmentation
 - Participated in 6 research projects leading to articles, including 3 at CVPR
- 2020–2021 **M2 in Applied Mathematics, Vision and Learning (Master MVA)**, *ENS Paris-Saclay*, Saclay.
Courses in Deep Learning, 3D Computer Vision, Convex and Discrete Optimization, Computational Statistics, Optimal Transport, Random Matrix Theory, Point Cloud and 3D Modeling and Kernel Methods.
- 2017–2021 **Master's Degree in Science and Executive engineering**, *Mines Paristech*, Paris.
Courses in Mathematics, Mechanics, Computer Science, Probability, Statistics, Physics, Automatics and Climate issues. Specialization in Robotics, Computer Vision and applied Mathematics. Projects in Deep Learning, Robotics, and 3D Reconstruction.
- 2018 **Research Exchange Semester**, *Massachusetts Institute of Technology*, Boston.
Research in Monocular SLAM, and robust SLAM in variable environments. Attended courses and conferences in Computer Vision and Robotics.
- 2015–2017 **Preparatory Class in Mathematics and Physics**, *Lycée Louis Pasteur*, Neuilly-sur-Seine.
Major courses in Mathematics and Physics, Minor courses in Computer Science and Engineering Sciences.
- 2015 **High School Diploma with high honors**, *Lycée Rocroy Saint Vincent de Paul*, Paris.

Experience

Professional Experience

- 2022–2025 **Research scientist**, *valeo.ai*, Paris, France.
Research position of my PhD, in the 3D team specialized on point clouds.
- 2021 **6-months Internship**, *valeo.ai*, Paris, France.
Research focused on self-supervision for point clouds, using contrastive methods between camera and LiDAR representations of a scene.
- 2020 **6-months Internship**, *IBM Research*, Tokyo, Japan.
Research among a team in IBM's Tokyo Research Lab. The aim was to study combination of automatic planners with Deep Learning to apply Logic Programming to difficult or noisy problem setups.
Detailed achievements:
 - Learned state-of-the-art in automatic planning and model-based reinforcement learning.
 - Contributed to IBM's intellectual property with multiple patent applications.
- 2019–2020 **6-months Internship**, *Scio*, London, United Kingdom.
Start-up specialized in automatic data extraction from unformatted documents. Built text and layout classifiers to improve parsing robustness. Carried out the transition from Keras to Pytorch for the main code.

Miscellaneous

- 2025–Today **Teaching Assistant**.
Giving courses and practicals on C++
- 2024 **Ellis Winter School on Foundation Models.**, *UvA*, Amsterdam, Netherlands.

2023–Today **Reviewer Duties.**

Reviewer for international conferences and journals including CVPR, ICCV, ECCV, 3DV, WACV and TPAMI
Outstanding reviewer at ECCV 2024 & CVPR 2025

2020 **School Project**, *Mines Paristech*, Paris, France.

3D Mesh reconstruction from Point Clouds, using Marching Cubes and implicit surface reconstruction

2019 **School Project**, *Mines Paristech*, Paris, France.

LIDAR point cloud classification algorithm using convolutional Neural Networks (U-Net).

2015–Today **Other.**

- Child Care for an autistic boy
- Personal Teacher for High School Students
- Management of an Organization, bringing fresh fruits and vegetables to students (20,000 € turnover)
- Instrumental practice of the Cello in an orchestra & of the Guitar
- Seasonal Employment & Blue-collar internship

Skills

Python Pytorch, Tensorflow, Numpy, Scipy, Regex, Cython and other standard libraries

C/C++ Standard libraries, OpenCV, Binding with Python

Other $\text{\LaTeX}$, Git, OpenOffice & Microsoft Office, Linux

Languages **French** (native), **English** (fluent), **German** (basic)

Publications

2025 **Clustering is back: Reaching state-of-the-art LiDAR instance segmentation without training**, *Under review*, [Corentin Sautier](#), Gilles Puy, Alexandre Boulch, Renaud Marlet, Vincent Lepetit.

Highlight: best method on the official SemanticKITTI Panoptic segmentation benchmark.

2025 **UNIT: Unsupervised Online Instance Segmentation through Time**, *3DV 2025*, [Corentin Sautier](#), Gilles Puy, Alexandre Boulch, Renaud Marlet, Vincent Lepetit.

Highlight: unsupervised instance segmentation on very long sequences.

2024 **Three Pillars improving Vision Foundation Model Distillation for Lidar**, *CVPR 2024*, Gilles Puy, Spyros Gidaris, Alexandre Boulch, Oriane Siméoni, [Corentin Sautier](#), Patrick Pérez, Andrei Bursuc, Renaud Marlet.

Highlight: best performing, yet simpler method for distilling image models in 3D networks.

2024 **BEVContrast: Self-Supervision in BEV Space for Automotive Lidar Point Clouds**, *3DV 2024*, [Corentin Sautier](#), Gilles Puy, Alexandre Boulch, Renaud Marlet, Vincent Lepetit.

Highlight: state-of-the-art self-supervised contrastive method for outdoor LiDAR point clouds.

2023 **ALSO: Automotive Lidar Self-supervision by Occupancy estimation**, *CVPR 2023*, Alexandre Boulch, [Corentin Sautier](#), Björn Michele, Gilles Puy, Renaud Marlet.

Highlight: best self-supervised reconstruction method using surface estimation.

2022 **Image-to-Lidar Self-Supervised Distillation for Autonomous Driving Data**, *CVPR 2022*, [Corentin Sautier](#), Gilles Puy, Spyros Gidaris, Alexandre Boulch, Andrei Bursuc, Renaud Marlet.

Highlight: method that created the field of image-to-LiDAR distillation.

2020 **State Prediction in TextWorld with a Predicate-Logic Pointer Network Architecture**, *IBM Research Tokyo*, [Corentin Sautier](#), Don Joven Agravante, Michiaki Tatsubori, KBRL at IJCAI-PRICAI 2020.

2020 **Towards Logical Model-based Reinforcement Learning: Lifted Operator Models**, *IBM Research Tokyo*, [Corentin Sautier](#), Don Joven Agravante, Michiaki Tatsubori, KBRL at IJCAI-PRICAI 2020.